2.52 Polar Coordinates: Cartesian Equations - Part 2

Worked Solutions

1 a)

$$r\cos\left(\theta-\frac{\pi}{3}\right)=1$$
$$r\cos\theta\cos\frac{\pi}{3}+r\sin\theta\sin\frac{\pi}{3}=1$$
$$x\cos\frac{\pi}{3}+y\sin\frac{\pi}{3}=1$$
$$\frac{1}{2}x+\frac{\sqrt{3}}{2}y=1$$
$$x+\sqrt{3}y=2$$

1 b)

$$r\cos\left(\theta+\frac{\pi}{6}\right)=1$$
$$r\cos\theta\cos\frac{\pi}{6}-r\sin\theta\sin\frac{\pi}{6}=1$$
$$x\cos\frac{\pi}{6}-y\sin\frac{\pi}{6}=1$$
$$\frac{\sqrt{3}}{2}x-\frac{1}{2}y=1$$
$$\sqrt{3}x-y=2$$

1 c)

$$r\sin\left(\theta+\frac{\pi}{4}\right)=1$$
$$r\sin\theta\cos\frac{\pi}{4}+r\cos\theta\sin\frac{\pi}{4}=1$$
$$y\cos\frac{\pi}{4}+x\sin\frac{\pi}{4}=1$$
$$\frac{\sqrt{2}}{2}y+\frac{\sqrt{2}}{2}x=1$$
$$\sqrt{2}x+\sqrt{2}y=2$$

1 d)

$$r\sin\left(\theta-\frac{\pi}{6}\right)=4$$
$$r\sin\theta\cos\frac{\pi}{6}-r\cos\theta\sin\frac{\pi}{6}=4$$
$$y\cos\frac{\pi}{6}-x\sin\frac{\pi}{6}=4$$

$$\frac{\sqrt{3}}{2}y - \frac{1}{2}x = 4$$
$$x - \sqrt{3}y = -8$$

2 a)

$$r = 3 + \cos 2\theta$$
$$r = 3 + \cos^2\theta - \sin^2\theta$$
$$r^3 = 3r^2 + r^2\cos^2\theta - r^2\sin^2\theta$$
$$(x^2+y^2)^{3/2} = 3x^2 + 3y^2 + x^2 - y^2$$
$$(x^2+y^2)^{3/2} = 4x^2 + 2y^2$$

2 b)

$$r^2 = \sin 2\theta$$
$$r^2 = 2\sin\theta\cos\theta$$
$$r^4 = 2r^2\sin\theta\cos\theta$$
$$(x^2+y^2)^2 = 2xy$$

2 c)

$$r^2 = 3\sec 2\theta$$
$$r^2\cos 2\theta = 3$$
$$r^2(\cos^2\theta - \sin^2\theta) = 3$$
$$r^2\cos^2\theta - r^2\sin^2\theta = 3$$
$$x^2 - y^2 = 3$$

2 d)

$$r^2 = 9\,\mathrm{cosec}\,2\theta$$
$$r^2\sin 2\theta = 9$$
$$r^2(2\sin\theta\cos\theta) = 9$$
$$2r\sin\theta r\cos\theta = 9$$
$$xy = \frac{9}{2}$$

2 e)

$$r = 6\cot\theta\operatorname{cosec}\theta$$
$$r = 6 \times \frac{\cos\theta}{\sin\theta} \times \frac{1}{\sin\theta}$$
$$r\sin^2\theta = 6\cos\theta$$
$$r^2\sin^2\theta = 6r\cos\theta$$
$$y^2 = 6x$$

3)

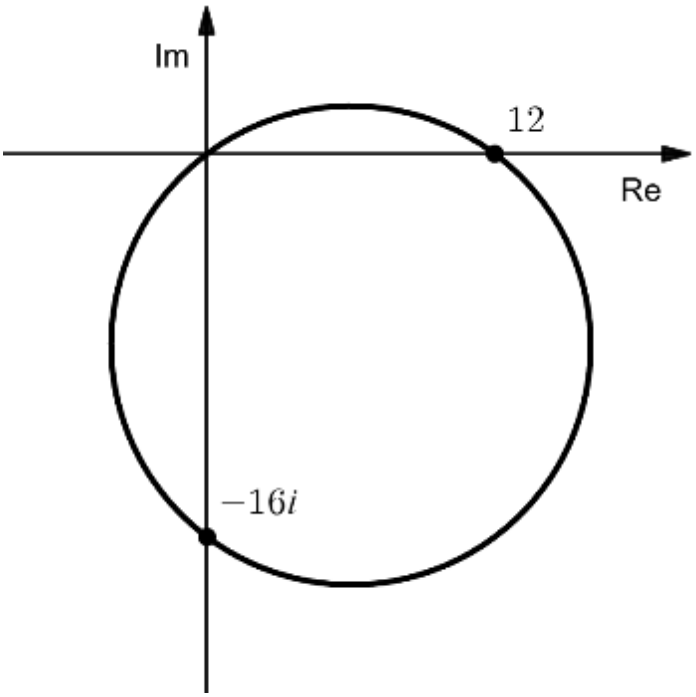
$$(x^2 + y^2)^2 = x^2 - y^2$$
$$r^4 = r^2\cos^2\theta - r^2\sin^2\theta$$
$$r^2 = \cos 2\theta$$
$$\therefore \mathrm{f}(\theta) = \cos 2\theta$$

4)

$$y = \frac{1}{x}$$
$$r\sin\theta = \frac{1}{r\cos\theta}$$
$$r^2 = \frac{1}{\sin\theta\cos\theta}$$
$$r^2 = \frac{2}{2\sin\theta\cos\theta}$$
$$r^2 = \frac{2}{\sin 2\theta}$$
$$r^2 = 2\operatorname{cosec} 2\theta$$
$$\therefore \mathrm{f}(\theta) = 2\operatorname{cosec} 2\theta$$

5 a)

Circle centred at $(6, -8)$ with radius 10

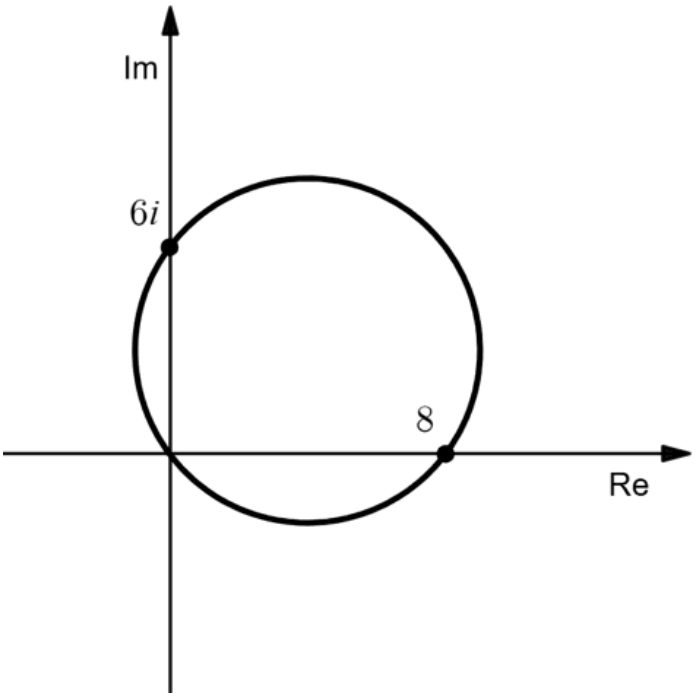


5 b)

$$(x-6)^2 + (y+8)^2 = 100$$
$$x^2 - 12x + 36 + y^2 + 16y + 64 = 100$$
$$x^2 - 12x + y^2 + 16y = 0$$
$$r^2\cos^2\theta - 12r\cos\theta + r^2\sin^2\theta + 16r\sin\theta = 0$$
$$r^2(\cos^2\theta + \sin^2\theta) - 12r\cos\theta + 16r\sin\theta = 0$$
$$r^2 = 12r\cos\theta - 16r\sin\theta$$
$$r = 12\cos\theta - 16\sin\theta \quad \text{as req.}$$

6 a)

Circle centred at $(4, 3)$ with radius 5



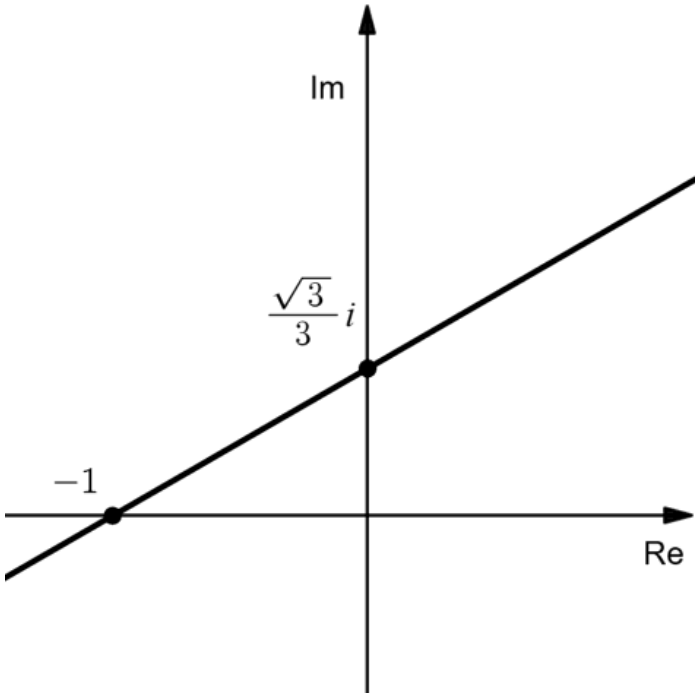
6 b)

$$(x-4)^2 + (y-3)^2 = 25$$
$$x^2 - 8x + 16 + y^2 - 6y + 9 = 25$$
$$x^2 - 8x + y^2 - 6y = 0$$
$$r^2\cos^2\theta - 8r\cos\theta + r^2\sin^2\theta - 6r\sin\theta = 0$$
$$r^2(\cos^2\theta + \sin^2\theta) - 8r\cos\theta - 6r\sin\theta = 0$$
$$r^2 = 8r\cos\theta + 6r\sin\theta$$
$$r = 8\cos\theta + 6\sin\theta \quad \text{as req.}$$

7 a)

$(1,0)$ and $(0,\sqrt{3})$ has midpoint $\left(\frac{1}{2},\frac{\sqrt{3}}{2}\right)$ and perpedicular gradient is $\frac{\sqrt{3}}{3}$

$$y-\frac{\sqrt{3}}{2}=\frac{\sqrt{3}}{3}\left(x-\frac{1}{2}\right)\Rightarrow y=\frac{\sqrt{3}}{3}x-\frac{\sqrt{3}}{6}+\frac{\sqrt{3}}{2}\therefore y=\frac{\sqrt{3}}{3}x+\frac{\sqrt{3}}{3}$$



7 b)

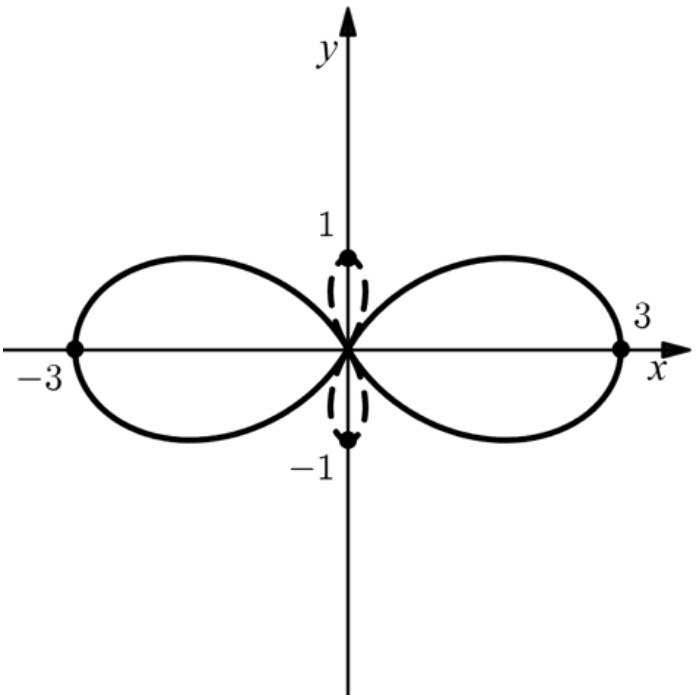
$$y=\frac{\sqrt{3}}{3}x+\frac{\sqrt{3}}{3}$$
$$r\sin\theta=\frac{\sqrt{3}}{3}r\cos\theta+\frac{\sqrt{3}}{3}$$
$$3r\sin\theta-\sqrt{3}r\cos\theta=\sqrt{3}$$
$$\sin(\theta-\alpha)=\sin\theta\cos\alpha-\cos\theta\sin\alpha$$

$\cos\alpha=3$ and $\sin\alpha=\sqrt{3}\Rightarrow R=\sqrt{12}$ and $\alpha=\arctan\left(\frac{\sqrt{3}}{3}\right)=\frac{\pi}{6}$

So that $r\sin\theta-\sqrt{3}r\cos\theta=\sqrt{3}$ becomes $\sqrt{12}r\sin\left(\theta-\frac{\pi}{6}\right)=\sqrt{3}$

$$\therefore r=\frac{1}{2}\operatorname{cosec}\left(\theta-\frac{\pi}{6}\right)\quad\text{as req.}$$

8 a)



8 b)

$$1 + 2\cos 2\theta = \sqrt{3}\sec\theta$$
$$1 + 2(2\cos^2\theta - 1) = \sqrt{3}\sec\theta$$
$$1 + 4\cos^2\theta - 2 = \sqrt{3}\sec\theta$$
$$4\cos^2\theta - 1 = \sqrt{3}\sec\theta$$
$$4\cos^3\theta - \cos\theta = \sqrt{3}$$
$$4\cos^3\theta - \cos\theta - \sqrt{3} = 0$$

9)

$\cosh^2 x - \sinh^2 x = 1 \Rightarrow 1 + \sinh^2 x = \cosh^2 x$

So that $r = \sqrt{1 + \sinh^2\theta} = \sqrt{\cosh^2\theta} = \cosh\theta$ since $r > 0$

For $y = x$ then $r\sin\theta = r\cos\theta$

So at the points of intersection: $\cosh\theta\sin\theta = \cosh\theta\cos\theta$

Since $\cosh\theta \neq 0$, then solve $\sin\theta = \cos\theta$

$$\Rightarrow \tan\theta = 1 \therefore \theta = \frac{\pi}{4}, \frac{5\pi}{4}, \ldots$$

For $\theta = \frac{\pi}{4}$:

$$r = \cosh\frac{\pi}{4} \Rightarrow x = r\cos\theta = \cosh\frac{\pi}{4}\cos\frac{\pi}{4} = \frac{\sqrt{2}}{2}\cosh\frac{\pi}{4}$$

and

$$y = r\sin\theta = \cosh\frac{\pi}{4}\sin\frac{\pi}{4} = \frac{\sqrt{2}}{2}\cosh\frac{\pi}{4}$$

So: $\left(\frac{\sqrt{2}}{2}\cosh\frac{\pi}{4}, \frac{\sqrt{2}}{2}\cosh\frac{\pi}{4}\right)$

For $\theta = \frac{5\pi}{4}$:

$$r = \cosh\frac{5\pi}{4} \Rightarrow x = r\cos\theta = \cosh\frac{5\pi}{4}\cos\frac{5\pi}{4} = -\frac{\sqrt{2}}{2}\cosh\frac{5\pi}{4}$$

and

$$y = r\sin\theta = \cosh\frac{5\pi}{4}\sin\frac{5\pi}{4} = -\frac{\sqrt{2}}{2}\cosh\frac{5\pi}{4}$$

So: $\left(-\frac{\sqrt{2}}{2}\cosh\frac{5\pi}{4}, -\frac{\sqrt{2}}{2}\cosh\frac{5\pi}{4}\right)$

$$\therefore \left(\frac{\sqrt{2}}{2}\cosh\frac{\pi}{4}, \frac{\sqrt{2}}{2}\cosh\frac{\pi}{4}\right) \text{ and } \left(-\frac{\sqrt{2}}{2}\cosh\frac{5\pi}{4}, -\frac{\sqrt{2}}{2}\cosh\frac{5\pi}{4}\right)$$

Or

$$\left(\frac{\sqrt{2}}{4}(e^{\pi/4} + e^{-\pi/4}), \frac{\sqrt{2}}{4}(e^{\pi/4} + e^{-\pi/4})\right)$$

and

$$\left(-\frac{\sqrt{2}}{4}(e^{5\pi/4} + e^{-5\pi/4}), -\frac{\sqrt{2}}{4}(e^{5\pi/4} + e^{-5\pi/4})\right)$$

10 a)

$$x^3 - y^3 = 3xy$$
$$r^3\cos^3\theta - r^3\sin^3\theta = 3r^2\cos\theta\sin\theta$$
$$r^3(\cos^3\theta - \sin^3\theta) = 3r^2\cos\theta\sin\theta$$

Since $r > 0$ then $\mathrm{f}(\theta) = \frac{3\cos\theta\sin\theta}{\cos^3\theta - \sin^3\theta}$

10 b)

Since $\cos\left(\frac{3\pi}{2} - \theta\right) = \cos\frac{3\pi}{2}\cos\theta + \sin\frac{3\pi}{2}\sin\theta = -\sin\theta$

and $\sin\left(\frac{3\pi}{2} - \theta\right) = \sin\frac{3\pi}{2}\cos\theta - \cos\frac{3\pi}{2}\sin\theta = -\cos\theta$ then:

$$\begin{aligned}\mathrm{f}\left(\frac{3\pi}{2} - \theta\right) &= \frac{3\cos\left(\frac{3\pi}{2} - \theta\right)\sin\left(\frac{3\pi}{2} - \theta\right)}{\cos^3\left(\frac{3\pi}{2} - \theta\right) - \sin^3\left(\frac{3\pi}{2} - \theta\right)} \\ &= \frac{3(-\sin\theta)(-\cos\theta)}{(-\sin\theta)^3 - (-\cos\theta)^3} = \frac{3\cos\theta\sin\theta}{-\sin^3\theta - -\cos^3\theta} \\ &= \frac{3\cos\theta\sin\theta}{\cos^3\theta - \sin^3\theta}\end{aligned}$$

10 c)

Since $\cos\left(\frac{3\pi}{2} - \theta\right) = -\sin\theta$ and $\sin\left(\frac{3\pi}{2} - \theta\right) = -\cos\theta$ this represents $(x, y) \mapsto (-y, -x)$ which is a reflection in the line $y = -x$

So the line of symmetry is $y = -x$

10 d)

When $\theta = \frac{\pi}{2}$, $\mathrm{f}\left(\frac{\pi}{2}\right) = \frac{3\cos\frac{\pi}{2}\sin\frac{\pi}{2}}{\cos^3\frac{\pi}{2} - \sin^3\frac{\pi}{2}} = 0$

When $\theta = \pi$, $\mathrm{f}(\pi) = \frac{3\cos\pi\sin\pi}{\cos^3\pi - \sin^3\pi} = 0$

So for $\frac{\pi}{2} \leq \theta \leq \pi$ it is symmetrical across $y = -x$ and starts and ends at the origin therefore it must be a loop.

11 a)

$$\begin{aligned}
13\cos^2\theta - 6\sqrt{3}\cos\theta\sin\theta + 7\sin^2\theta &= 13\cos^2\theta - 3\sqrt{3}\sin 2\theta + 7\sin^2\theta \\
&= 13\cos^2\theta - 3\sqrt{3}\sin 2\theta + 7(1 - \cos^2\theta) \\
&= 6\cos^2\theta - 3\sqrt{3}\sin 2\theta + 7 \\
&= 6\left(\frac{1}{2} + \frac{1}{2}\cos 2\theta\right) - 3\sqrt{3}\sin 2\theta + 7 \\
&= 3 + 3\cos 2\theta - 3\sqrt{3}\sin 2\theta + 7 \\
&= 10 + 3\cos 2\theta - 3\sqrt{3}\sin 2\theta \quad \text{as req.}
\end{aligned}$$

11 b)

$$\begin{gathered}
13x^2 - 6\sqrt{3}xy + 7y^2 = 1 \\
13r^2\cos^2\theta - 6\sqrt{3}r^2\cos\theta\sin\theta + 7r^2\sin^2\theta = 1 \\
r^2(13\cos^2\theta - 6\sqrt{3}\cos\theta\sin\theta + 7\sin^2\theta) = 1 \\
r^2 = \frac{1}{13\cos^2\theta - 6\sqrt{3}\cos\theta\sin\theta + 7\sin^2\theta}
\end{gathered}$$

Since $13\cos^2\theta - 6\sqrt{3}\cos\theta\sin\theta + 7\sin^2\theta = 10 + 3\cos 2\theta - 3\sqrt{3}\sin 2\theta$ then

$$r^2 = \frac{1}{10 + 3\cos 2\theta - 3\sqrt{3}\sin 2\theta}$$

Consider $3\cos 2\theta - 3\sqrt{3}\sin 2\theta$ and so

$$R\sin(2\theta + \alpha) = R\sin 2\theta\cos\alpha + R\cos 2\theta\sin\alpha$$

$R = \sqrt{3^2 + (3\sqrt{3})^2} = 6$ and $\alpha = \arctan\left(\frac{3}{3\sqrt{3}}\right) = \frac{\pi}{6}$

So that

$$r^2 = \frac{1}{10 - 6\sin(2\theta - \frac{\pi}{6})}$$

$\therefore$ max and min of r^2 are $r^2 = \frac{1}{16}$ and $r^2 = \frac{1}{4}$

So the greatest distance is $\frac{1}{2}$ and the least distance is $\frac{1}{4}$